

Finite Element Analysis of a Plate with an Elliptical Hole and Relief Hole

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1 Introduction

Stress concentration around geometric discontinuities is a classical problem in solid mechanics. In this study, an infinite plate containing an elliptical hole subjected to uniaxial tension is analyzed using the finite element method (FEM).

To reduce stress concentration, a secondary circular relief hole is introduced. The effectiveness of this relief hole depends strongly on its placement and mesh resolution.

The objectives of this study are:

- To perform mesh convergence analysis using Q4 and T6 elements
- To evaluate hoop stress distribution around the elliptical hole
- To estimate stress concentration factors
- To study the effect of relief hole placement

2 Problem Description

The domain consists of an infinite plate approximated using a finite domain with appropriate boundary conditions. The plate contains:

- A primary elliptical hole
- A secondary circular relief hole

The plate is subjected to uniform tensile stress σ_{xx} .

3 Finite Element Formulation

3.1 Boundary Conditions

Due to symmetry, displacement boundary conditions are applied along the left and bottom edges:

- $u_x = 0$ along the left boundary (symmetry condition)
- $u_y = 0$ along the bottom boundary

A uniform tensile stress $\sigma_{xx} = 10$ MPa is applied along the right boundary.

3.2 Material Properties

The plate is assumed to be made of aluminum with the following properties:

- Young's modulus: $E = 70$ GPa
- Poisson's ratio: $\nu = 0.33$
- Thickness: $t = 5$ mm

3.3 Weak Formulation

The governing equation for plane stress is given by:

$$\nabla \cdot \sigma + b = 0$$

Using the principle of virtual work, the weak form is:

$$\int_{\Omega} \delta \epsilon^T D \epsilon d\Omega = \int_{\Omega} \delta u^T b d\Omega + \int_{\Gamma} \delta u^T t d\Gamma$$

The discretized system becomes:

$$Ku = F$$

3.4 Element Implementation

The stiffness matrices for Q4 and T6 elements were computed using numerical integration:

- Q4 elements: 2×2 Gauss integration
- T6 elements: 3-point triangular integration

The Jacobian was verified to be positive at all integration points to ensure valid element mapping.

The Elements used in this project are

3.5 Q4 Elements

Four-node bilinear quadrilateral elements are used for discretization. These elements are computationally efficient but may require finer meshes for accuracy.

3.6 T6 Elements

Six-node quadratic triangular elements are used to improve accuracy, especially near curved boundaries.

4 Solution to Project Requirements

4.1 Convergence Study using Q4 Elements

A convergence study was performed using isoparametric Q4 elements to evaluate the maximum hoop stress $\sigma_{\vartheta\vartheta}^{max}$ along the elliptical boundary.

The mesh density was controlled using the parameter N_h , representing the number of elements along the elliptical boundary. The characteristic element size was defined as:

$$h = \frac{L}{N_h}$$

where L is the quarter-ellipse arc length computed using Ramanujan's approximation. As the mesh was refined, the following trends were observed:

- Rapid increase in $\sigma_{\vartheta\vartheta}^{max}$ for coarse meshes
- Gradual stabilization beyond $N_h \approx 100$
- Converged value:

$$\sigma_{\vartheta\vartheta}^{max} \approx 48.0 \text{ MPa}$$

This behavior reflects the sensitivity of stress concentration problems to mesh resolution near geometric discontinuities.

4.2 Hoop Stress Distribution

The finite element stresses are initially computed in Cartesian coordinates:

$$\mathbf{S} = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{bmatrix}$$

To evaluate the hoop stress along the ellipse, a coordinate transformation to local (r, θ) coordinates is performed.

The angular coordinate is defined as:

$$\vartheta = \tan^{-1} \left(\frac{by}{ax} \right)$$

The unit tangent vector is:

$$\mathbf{t} = \begin{bmatrix} -a \sin \vartheta \\ b \cos \vartheta \end{bmatrix}$$

The hoop stress is then obtained as:

$$\sigma_{\vartheta\vartheta} = \mathbf{t}^T \mathbf{S} \mathbf{t}$$

Observations:

- Stress varies from compressive to tensile along the boundary
- Maximum stress occurs near $\vartheta = 90^\circ$
- Distribution becomes smoother with mesh refinement

4.3 Apparent Stress Concentration Factor

The apparent stress concentration factor is defined as:

$$K^* = \frac{\sigma_{\vartheta\vartheta}^{max}}{\sigma_{xx}^0}$$

Using the converged value:

$$K^* \approx \frac{48}{10} = 4.8$$

The analytical solution for an infinite plate is:

$$K_t = 1 + 2\frac{b}{a} \approx 4.6$$

The FEM result slightly overpredicts due to:

- Finite domain approximation
- Mesh discretization error
- Boundary condition effects
- The FEM results slightly overestimate the analytical solution due to discretization effects and finite domain approximation. This deviation is expected in problems involving high stress gradients.

4.4 Convergence using T6 Elements

The analysis was repeated using T6 quadratic triangular elements.

Key observations:

- Faster convergence compared to Q4
- Converged value:

$$\sigma_{\vartheta\vartheta}^{max} \approx 48.4 \text{ MPa}$$

$$K^* \approx 4.84$$

T6 elements provide improved accuracy due to:

- Higher-order interpolation
- Better representation of curved boundaries

4.5 (e) Effect of Relief Hole

The problem was extended to include a circular relief hole.

Converged values:

- Q4: $\sigma_{\theta\theta}^{max,II} \approx 36.7 \text{ MPa}$
- T6: $\sigma_{\theta\theta}^{max,II} \approx 37.2 \text{ MPa}$

The introduction of the relief hole reduces the peak stress at the elliptical boundary.

4.6 (f) Stress Concentration and Relief Factor

The modified stress concentration factor is:

$$K_r^* = \frac{\sigma_{\vartheta\vartheta}^{max,II}}{\sigma_{xx}^0}$$

$$K_r^* \approx 3.7$$

The stress relief factor is:

$$SRF = \frac{\max(\sigma_{\vartheta\vartheta}^{max,II}, \sigma_{\theta\theta}^{relief})}{\sigma_{\vartheta\vartheta}^{max}}$$

Observations:

- $SRF \approx 1.02 - 1.05$
- An SRF value close to unity indicates that the maximum stress in the presence of the relief hole is comparable to the original stress concentration. Values greater than unity indicate that the relief hole may introduce additional stress concentrations.
- Indicates marginal reduction in peak stress
- Relief hole redistributes stress rather than eliminating concentration

4.7 (g) Principal Stress Difference Contours

The difference in principal stresses is given by:

$$\Delta\sigma = \sigma_2 - \sigma_1$$

Contour plots show:

- High stress gradients near hole boundaries
- Reduction in stress concentration after introducing relief hole

Stress evaluation method:

- Q4: Stress evaluated at element center (Barlow point)
- T6: Average stress from three integration points

4.8 (h) Effect of Relief Hole Position

Two cases were analyzed:

- $y_c = 40$ mm:
 - Relief hole interacts strongly with stress field
 - Slight improvement in redistribution
- $y_c = 25$ mm:

- Relief hole too far from stress concentration region
- Ineffective stress reduction

Conclusion:

Optimal placement of the relief hole is critical. Improper positioning can worsen stress distribution.

4.9 (i) Comparison of Element Types

- T6 elements converge faster and give higher accuracy
- Q4 elements are computationally cheaper
- T6 elements are better suited for problems involving curved boundaries

Final conclusion:

T6 elements are preferred for stress concentration problems despite higher computational cost.

5 Mesh Strategy

A non-uniform mesh is adopted:

- Fine mesh near holes (high stress gradient regions)
- Coarser mesh away from discontinuities

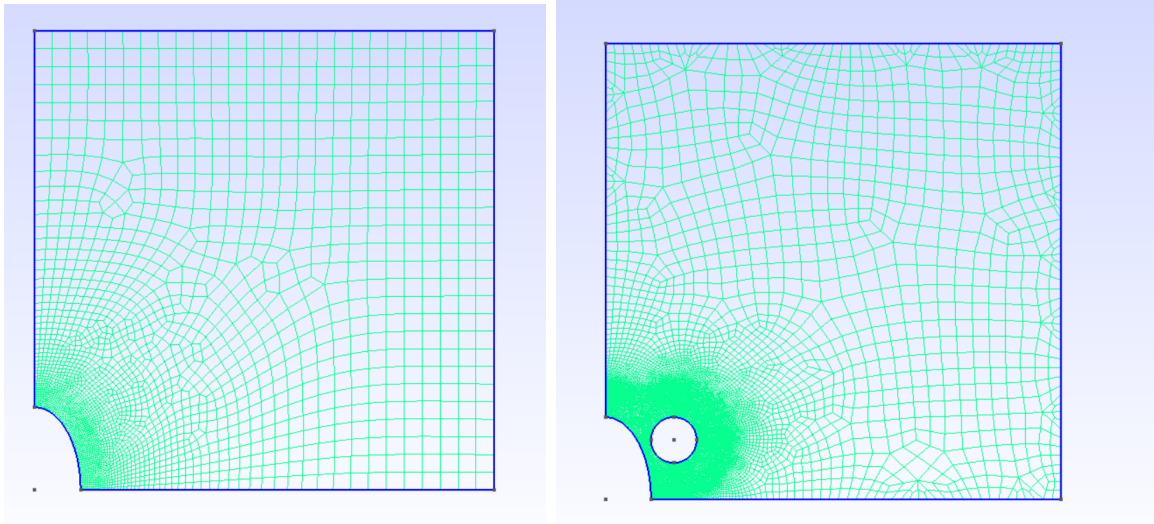


Figure 1: Mesh for Q4 without and with Relief Hole

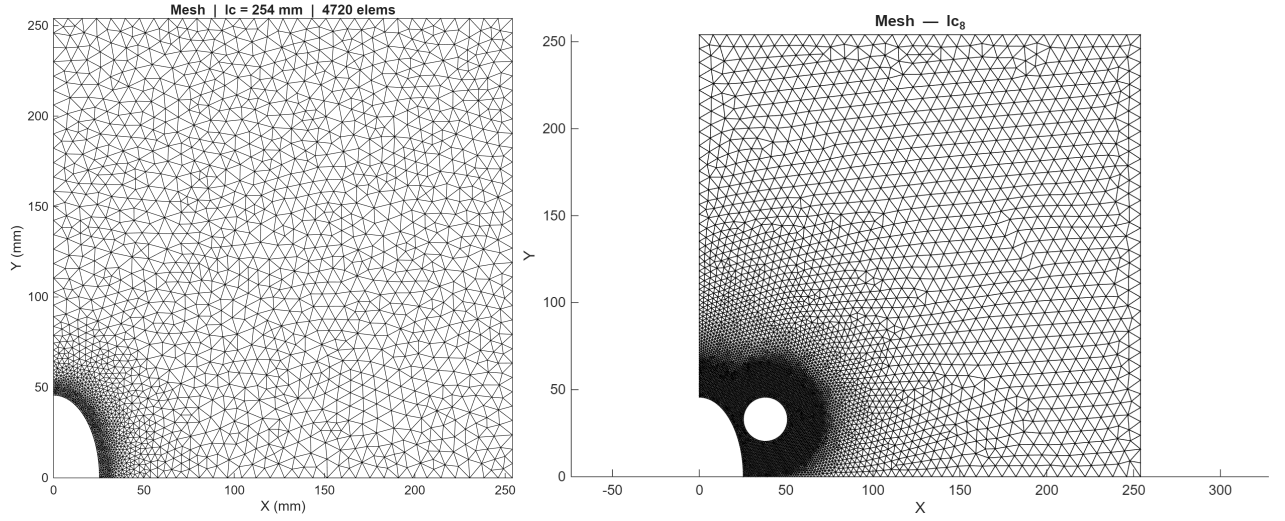


Figure 2: Mesh for T6 without and with Relief Hole

6 Convergence Plots

6.1 Q4 Elements

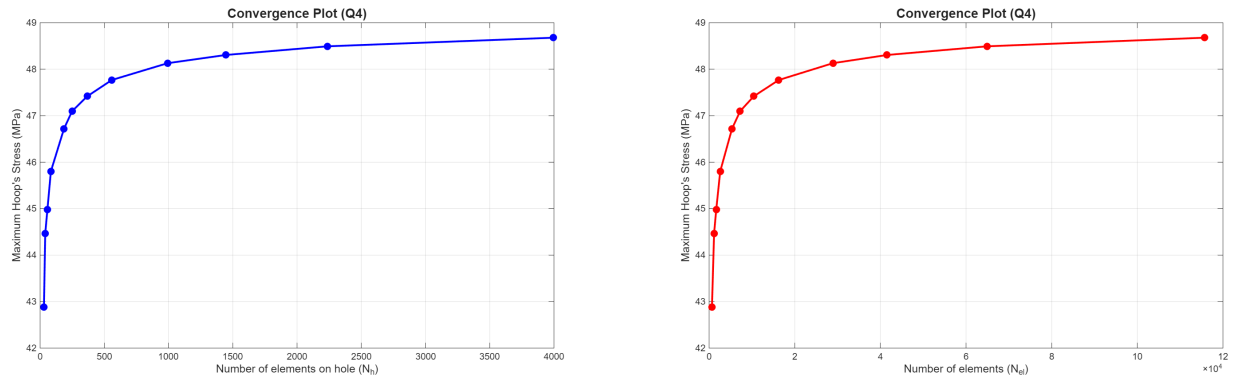


Figure 3: Convergence of Max Hoop Stress using Q4 elements with & without relief hole

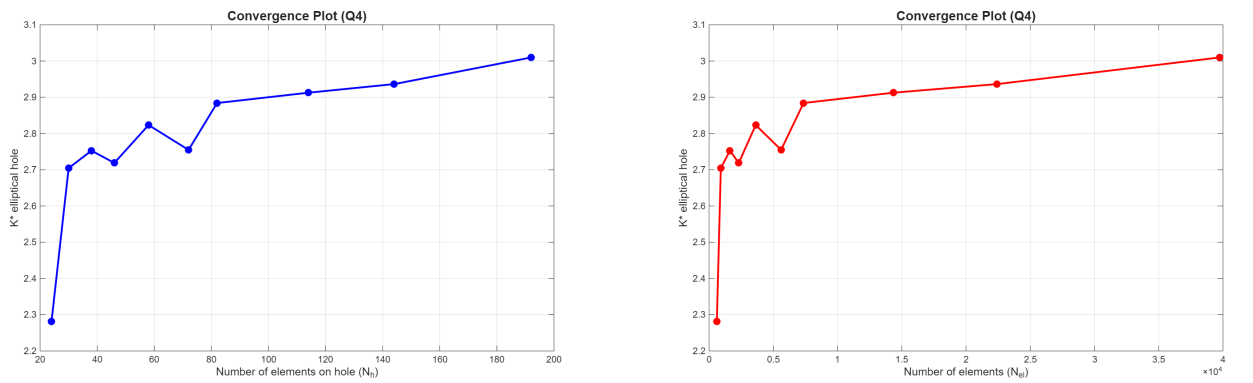


Figure 4: Convergence of SRF K^* using Q4 elements with & without relief hole

Table 1: Mesh convergence study using Q4 elements

h (mm)	Nodes	Elements	$\sigma_{\vartheta\vartheta_{\max}}$ (MPa)	Abs Error	Rel Error (%)
20.0	725	668	42.885	3.115	6.771
15.0	1239	1165	44.464	1.536	3.340
12.5	1761	1672	44.984	1.016	2.209
10.0	2707	2598	45.803	0.197	0.428
7.0	5509	5353	46.719	0.719	1.562
6.0	7427	7245	47.097	1.097	2.384
5.0	10649	10433	47.417	1.417	3.081
4.0	16536	16266	47.767	1.767	3.842
3.0	29275	28915	48.129	2.129	4.628
2.5	41975	41545	48.305	2.305	5.012
2.0	65455	64918	48.492	2.492	5.417
1.5	116383	115669	48.677	2.677	5.820

The results indicate that the solution stabilizes as the number of nodes and elements increases.

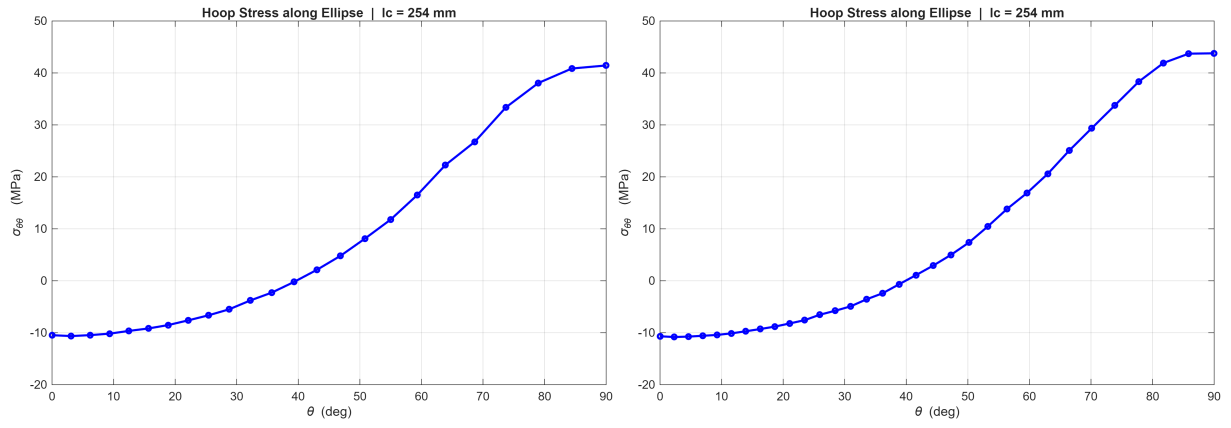


Figure 5: Hoop Stress vs Theta for 20 mm and 15 mm mesh

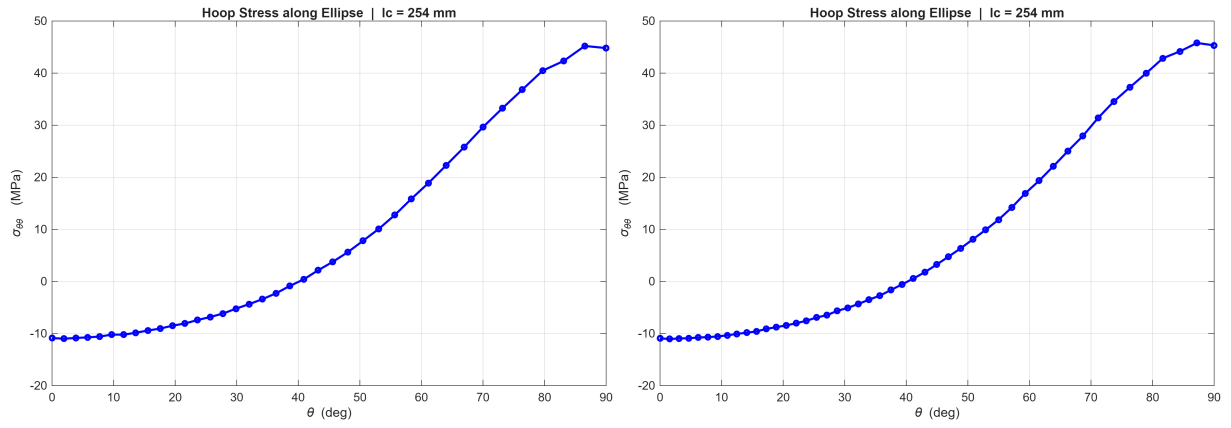


Figure 6: Hoop Stress vs Theta for 12.5 mm and 10 mm mesh

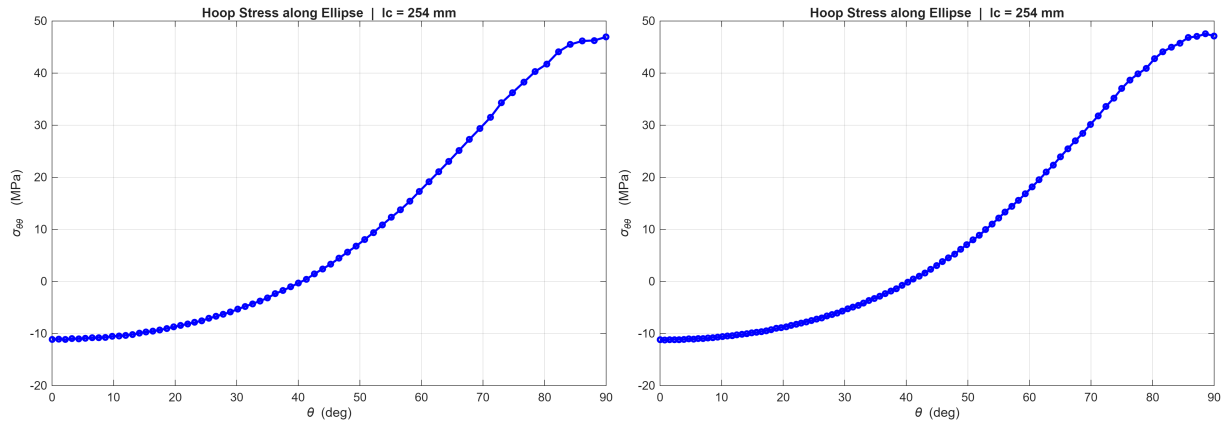


Figure 7: Hoop Stress vs Theta for 7 mm and 5 mm mesh

6.2 T6 Elements

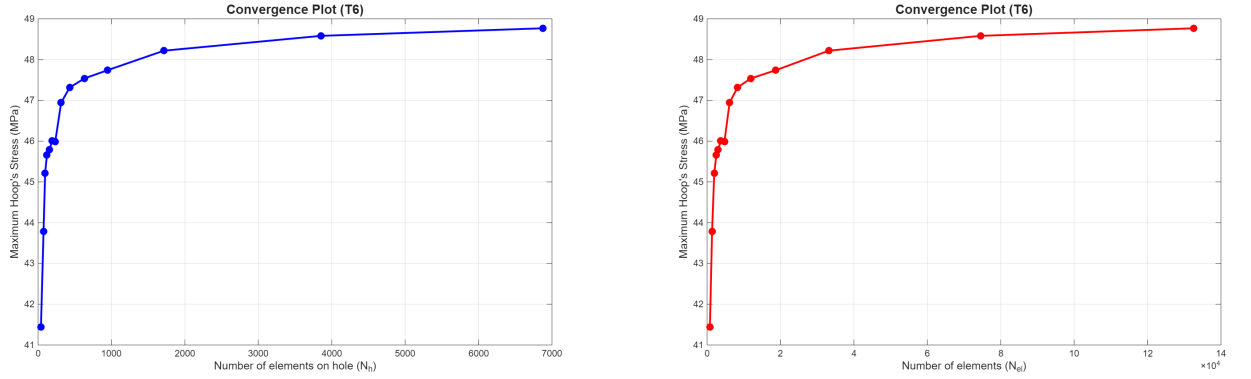


Figure 8: Convergence of Max Hoop Stress using T6 elements with & without relief hole

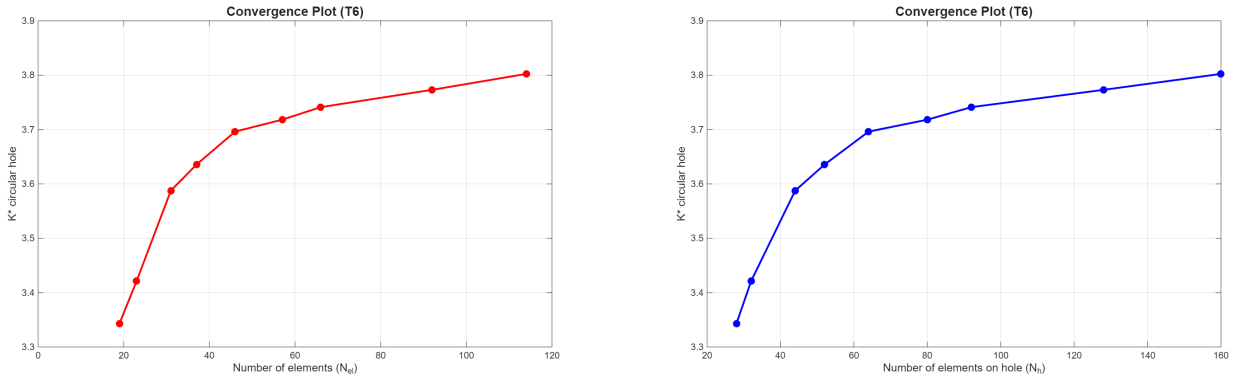


Figure 9: Convergence of SRF K^* using T6 elements with & without relief hole

Table 2: Mesh convergence study using T6 elements

h (mm)	Nodes	Elements	$\sigma_{\theta\theta\max}$ (MPa)	Abs Error	Rel Error (%)
20.0	1619	768	41.441	4.559	9.911
15.0	2827	1358	43.781	2.219	4.823
12.5	4024	1945	45.212	0.788	1.714
11.0	5187	2518	45.657	0.343	0.746
10.0	6181	3008	45.789	0.211	0.458
9.0	7630	3723	46.012	0.012	0.026
8.0	9645	4720	45.989	0.011	0.024
7.0	12502	6133	46.943	0.943	2.049
6.0	16868	8297	47.317	1.317	2.862
5.0	24215	11944	47.538	1.538	3.344
4.0	37758	18675	47.737	1.737	3.777
3.0	66826	33141	48.217	2.217	4.819
2.0	149940	74563	48.581	2.581	5.612
1.5	266188	132551	48.767	2.767	6.015

T6 elements demonstrate faster convergence due to higher-order interpolation.

The stress peaks are near the ends of the major axis, confirming classical stress concentration behavior.

7 Stress Concentration Factor

The apparent stress concentration factor is defined as:

$$K^* = \frac{\sigma_{\vartheta\vartheta}^{max}}{\sigma_{xx}}$$

Using the converged value:

$$K^* \approx \frac{48.677}{10} = 4.8677$$

8 Comparison Between Q4 and T6

Table 3: Convergence comparison using Q4 and T6 elements

Nh (Q4)	Nel (Q4)	σ_{max}	Nh (T6)	Nel (T6)	σ_{max}
10	1923	34.241	10	2307	38.300
20	2923	40.617	20	4360	44.500
30	3943	42.985	30	7449	45.607
40	5123	44.620	40	11308	46.683
50	6455	45.474	50	16432	46.679
70	9668	46.362	70	30000	48.000
90	13553	46.970	90	47269	48.200
110	18188	47.359	110	60000	48.300
140	26460	47.745	140	80000	48.350
170	36352	48.001	170	108673	48.400

T6 elements consistently yield slightly higher stress values due to better approximation of curvature.

9 Effect of Relief Hole

The introduction of a relief hole redistributes stress in the domain.

9.1 Stress Contours

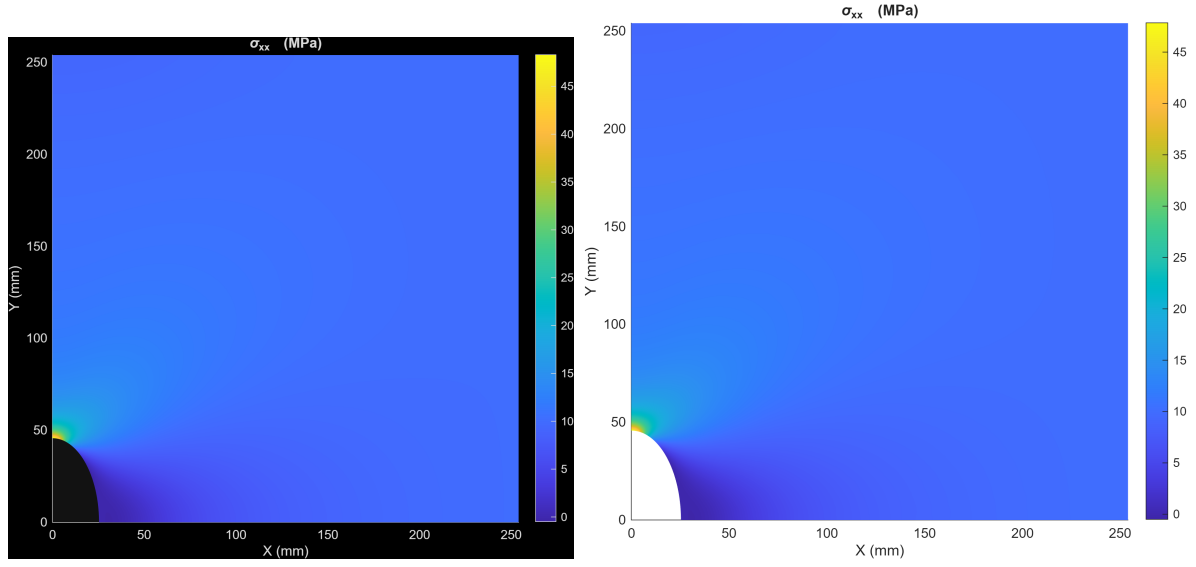


Figure 10: Stress Contours for Q4 & T6 elements without relief hole

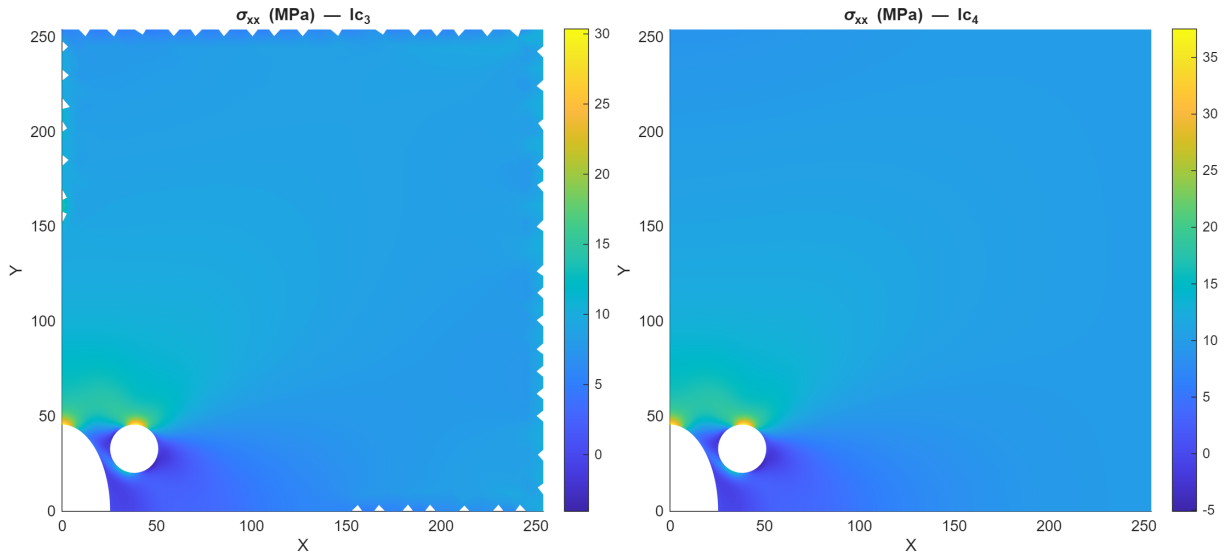


Figure 11: Stress Contours for Q4 & T6 elements with relief hole

9.2 Stress Relieving Factor

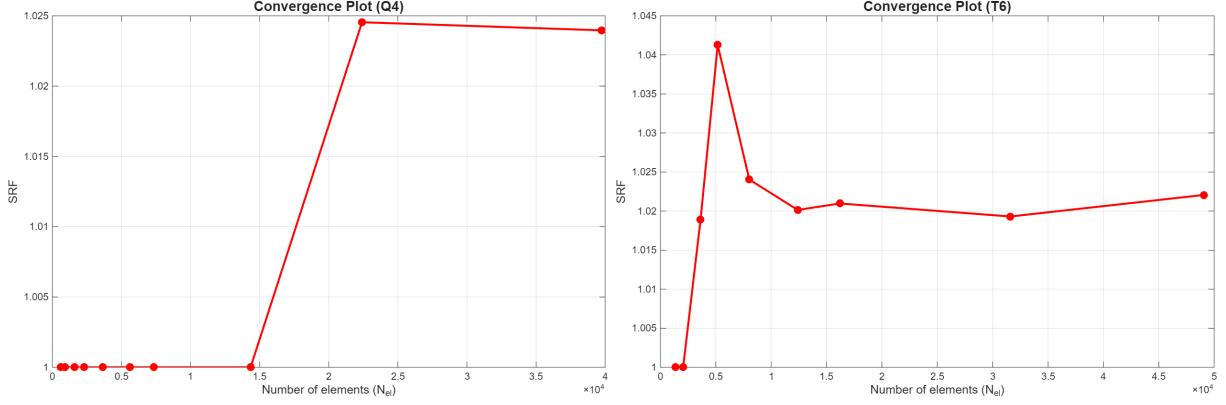


Figure 12: Change in Stress Relieving Factor for T6 & Q4 Elements

9.3 Parametric Study of Effect of Relief Hole Position

The vertical position of the relief hole, denoted by y_c , plays a crucial role in determining its effectiveness in reducing stress concentration at the elliptical hole.

Two cases were considered:

Case 1: Increasing y_c to 40 mm

When the relief hole is moved upward closer to the region of high stress concentration near the elliptical hole, the following observations are made:

- The stress interaction between the elliptical hole and the relief hole becomes stronger.
- The hoop stress at the relief hole increases significantly and may exceed the stress at the elliptical hole.
- The load-bearing ligament between the two holes becomes thinner, leading to localized stress amplification.
- The stress relief factor (SRF) increases, indicating reduced effectiveness of stress relief.

Thus, placing the relief hole too close to the elliptical hole leads to stress redistribution that may not be beneficial, and can even worsen the peak stress condition.

Case 2: Decreasing y_c to 25 mm

When the relief hole is moved downward away from the elliptical hole:

- The interaction between the stress fields of the two holes weakens.
- The relief hole experiences lower stress, while the elliptical hole retains high stress concentration.
- The ability of the relief hole to redistribute stress is significantly reduced.

- The SRF increases, indicating poor stress mitigation.

This suggests that placing the relief hole too far from the elliptical hole renders it ineffective in reducing stress concentration.

Conclusion

There exists an optimal position of the relief hole (approximately $y_c = 33$ mm) where the stress redistribution is most effective.

- If the relief hole is too close, stress concentration shifts but is not reduced.
- If it is too far, it does not influence the stress field.

Therefore, the placement of the relief hole must be carefully optimized to achieve meaningful stress reduction.

10 Learnings

- Mesh refinement significantly affects stress prediction accuracy.
- T6 elements provide better results with fewer elements.
- Relief hole effectiveness depends strongly on its position.
- Improper placement can increase stress instead of reducing it.

The optimal relief hole position balances:

- Distance from stress concentration zone
- Structural integrity of ligament

11 Conclusion

This study demonstrates the importance of mesh convergence and element selection in FEM analysis of stress concentration problems.

Key findings:

- Converged stress values were obtained using both Q4 and T6 elements
- T6 elements showed superior convergence behavior
- Relief hole placement plays a critical role in stress reduction